

Name:

Abdul-Rahman

Reg no:

241854 - CS4A

Course
MVC

Assignment - 3

⇒ Q1) $h(x,y) = 4x^2 + y^2 - 4x$ at $(1,2)$

a) ⇒ partial derivative

$$\frac{\partial h}{\partial x} = 8x - 4$$

$$\frac{\partial h}{\partial y} = 2y$$

at $(1,2)$

$$\frac{\partial h}{\partial x} = 4$$

$$\frac{\partial h}{\partial y} = 4$$

height increase by factor of 4

b) ⇒ $\Delta h(1,2) = \langle 4, 4 \rangle$

$$|\Delta h| = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$$

at 45° , direction of fastest increase.

c) ⇒

Since both ∂x & ∂y are equal 4, the

increase is equal in both directions.

$$\Rightarrow Q2) \quad x = t^2, \quad y = t + 1$$

chain rule:

$$\frac{dT}{dt} = (\partial T / \partial x) \times (dx/dt) + (\partial T / \partial y) \times (dy/dt)$$

$$\frac{dx}{dt} = 2t \quad \Bigg| \quad \frac{dy}{dt} = 1 \quad \Bigg| \quad T_x, T_y$$

$$b) \Rightarrow \text{At } T=1$$

$$\begin{aligned} \frac{dT}{dt} &= 2 * T_x + 1 * T_y \\ &= 2(1) + 1(1) \\ &= 3 \end{aligned}$$

small positive value indicate slow increase

c) A rising temperature is the indication of hot place, exhaust output, fire area or danger zone.

$$\Rightarrow Q3) \quad E(x, y) = x^2 + y^2 - 6x - 2y + 10$$

a) optimal position:

$$\partial E / \partial x = 2x - 6 = 0 \Rightarrow x = 3$$

$$\partial E / \partial y = 2y - 2 = 0 \Rightarrow y = 1$$

Critical point (3, 1) $\Rightarrow$ optimal position.

b) local or global optimum:

$$\partial^2 E / \partial x^2 = 2, \quad \partial^2 E / \partial y^2 = 2$$

$$\partial^2 E / \partial x \partial y = 0$$

$$D = (2)(2) - (0)^2 = 4 > 0, \text{ global minimum.}$$

E is paraboloid opening upwards.

c) AI Strategy:

Since $\bar{E}$ has one critical point
& is global minimum, gradient movements
will converge to $(3, 1)$. AI doesn't
take risk in false local optima.

$$\Rightarrow \text{Q4) } R(x, y) = x^2 + y^2 - 2x \text{ on } 0 \leq x \leq 3, 0 \leq y \leq 3$$

a) Interior critical points:

$$\partial R / \partial x = 2x - 2 = 0 \Rightarrow x = 1$$

$$\partial R / \partial y = 2y = 0 \Rightarrow y = 0$$

No interior critical point exist!

b) Check boundaries:

$$y=0: R(x,0) = x^2 - 2x \Rightarrow \text{min at } x=1: (R(1,0) = -1 \\ \text{endpoints: } R(0,0) = 0, R(3,0) = 3$$

$$y=3: R(x,3) = x^2 + 9 - 2x \Rightarrow \text{min at } x=1: R(1,3) = 8 \\ \text{endpoints: } R(0,3) = 9, R(3,3) = 12$$

$$x=0: R(0,y) \Rightarrow \text{range } [0, 9] \text{ from } y=0 \text{ to } y=3$$

$$x=3: R(3,y) \Rightarrow \text{range } [3, 12] \text{ from } y=0 \text{ to } y=3$$

$$\text{minimum: } R(1,0) = -1, \text{ maximum: } R(3,3) = 12$$

c)

players should go $(3,3) \Rightarrow R = 12$
maximum reward

$$\Rightarrow Q5) f(x,y) = 4x + 3y \text{ on } x^2 + y^2 = 9$$

use Lagrange multipliers: $\nabla f = \lambda \nabla g$
where $g(x,y) = x^2 + y^2 = 9$

$$\nabla f = (4, 3) \neq \nabla g = (2x, 2y)$$

$$4 = 2\lambda x \Rightarrow \lambda = 2/x$$

$$3 = 2\lambda y \Rightarrow y = 3/2\lambda$$

Substitute into g :

$$(2/x)^2 + (3(2/x))^2 = 9$$

$$4/x + 9/4x^2 = 9$$

$$16(4x^2) = 9$$

$$\Rightarrow x = \pm 5/6$$

for maximum:

$$x = 2/(5/6) = 12/5$$

$$y = 3(2 \times (5/6)) = 9/5$$

$$f(12/5, 9/5) = 4(12/5) + 3(9/5) = 75/5 = 15$$

maximum score = 15

b) At optimal point gradient is exactly parallel to gradient of $g(x, y)$, Means curve of f is tangent to g . So any movements along circle decrease f .